Adversarial review and revision: methods

How version 3 was reviewed from six perspectives, how its counts were audited, and how version 4 was derived

Centre for Enterprise Innovation (CEI) AI Robotics project, with Claude Code

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1. Purpose

The study's claims rest on a model-labelled collection assembled in two rounds with different selection mechanisms. Neither the labels nor the counts had been checked by anyone outside the pipeline that produced them. Version 4 therefore subjected version 3 to a structured adversarial review before any further claim was made, and treated the review as a measurement exercise: each reviewer was asked to find what was wrong and to show it with recomputed numbers or document identifiers, and the findings were converted into rules, corrections, additional measurements and rewritten claims. This section describes the procedure, the evidence it produced and its limits.

2. The six reviews

Six reviews were commissioned in parallel from independent reviewers, each an AI agent (Claude) with a written brief and no access to the others' work. The perspectives were chosen to cover the kinds of error a quantitative content analysis can contain: a data scientist and applied statistician (definitions, denominators, selection effects, tests and models); an AI researcher specialising in LLM annotation (validity of the labels, reliability evidence, prompt inputs, reproducibility); a conference reviewer in the ICLR/NeurIPS style (soundness of design, novelty, presentation, availability); a technology and policy journalist (whether the findings are understandable, trustworthy and quotable for a general audience); an AI and robotics governance expert (whether the instruments and their classifications are right, and what is missing); and a data-visualisation and dashboard specialist (whether every figure says what its data say, and whether the dashboard works).

Each brief named the files to read in order of importance, listed the earlier rounds' fixes with the instruction to check whether they had held, gave a focus list of specific risks, and prescribed the deliverable: a verdict, a findings table with severity, location, problem, evidence and suggested fix, and a ranked short list of changes. Reviewers were told to recompute any number they doubted from the master table using the study's own set definitions, and the AI researcher and governance expert were asked to read documents and model inputs themselves. Reviews ran for 14 to 25 minutes each; findings tables ranged from about a dozen entries (data scientist, journalist) to about sixty (AI researcher, visualisation

reviewer), the conference reviewer wrote prose weaknesses with scores, the governance expert's classification audit inspected 159 instruments (165 table rows) and found at least one error in 68% and a material error in 62% of a risk-stratified, not random, sample, and the visualisation reviewer saved 39 screenshots.

3. What the reviews found

The six perspectives converged on four structural problems, which is the basis on which they were treated as defects of the study rather than matters of taste.

First, the counting unit. Version 3 counted every text of instrument class as one instrument, so commencement orders, levy acts, individual exemptions and approvals, EEA incorporation decisions, language-version corrigenda, definitions sections, register pages and metadata stubs appeared as binding instruments beside the acts they belong to, and yearly renewals and successive versions of one regime appeared several times. Second, the collection round as a confound. The embodied-versus-general comparisons pooled round-1 documents (web-collected research and policy texts) with round-2 documents (database-harvested legal texts of a different genre, length and language mix), and the covariate adjustment did not include the round; the headline "ethical core deficit" of version 3 was shown not to exist within round 1. Third, the excerpt cleaner. A rule introduced in version 2 to strip navigation lines from the top of web captures deleted lines without whitespace-delimited words or ASCII sentence punctuation, which removed most of the text of Chinese, Japanese and Korean documents before the model saw it; a median of a quarter of each Chinese document reached the model and 83 Chinese documents naming robots or AI were labelled as not about AI. Fourth, capture tier. Statistics about what standards and binding instruments specify "in depth" included catalogue scope pages and stubs, contrary to the study's own rule.

The reviews also recorded the regressions of earlier fixes at scale in round 2 (enacted proposals still "under consideration", draft standards counted as published, mandatory Chinese standards as soft law, position papers as draft instruments), the absence of any reliability evidence for round-2 labels, three different denominators for one concept, a dashboard methods panel that still described version 1, and figure titles that their own data contradicted.

4. Additional measurements

Two measurements were commissioned from the findings before any claim was rewritten.

A precision audit drew a simple random sample of 100 of the 299 version-3 "binding instruments reaching embodied AI" (seeded, documented draw) and had an independent agent read each text and judge five criteria: whether the capture is a distinct principal instrument or substantive amendment, whether it is binding, whether it contains substantive provisions specific to an embodied system, whether it contains any autonomy or AI provision, and whether it names AI explicitly. The shares were 67%, 97%, 56%, 67% and 14% (95% Wilson intervals of about plus or minus ten points); 54% (44 to 63) met the first three jointly, implying about 161 (132 to 190) genuine instruments in the set of 299. The audit also recorded, for each document, whether it belongs to a family of related captures.

A round-2 reliability sample of 160 documents, stratified over standards, EU legal texts, East Asian texts, intergovernmental documents, United States texts and other national texts, was

re-annotated under the study's blind extended-excerpt policy by a second model (Claude Opus 5.5) and by a same-model re-run (Claude Sonnet 5.5). Against the final labels, median Cohen's kappa over single and ordinal fields was 0.86 for the second model and 0.91 for the re-run, comparable to the round-1 values of 0.85 and 0.91; the weakest fields were page quality, stance toward regulation, publication stage, date basis and instrument status. Unweighted and level-2 (treated in depth) variants were computed for all six reliability comparisons.

5. From findings to version 4

Findings were triaged into five kinds of change, each with a documented implementation.

Admission and unit rules, applied to every document by stated regular expressions on titles and thresholds on capture fields, assign each text of instrument class to a unit: principal instrument, amendment, ancillary (commencement, levy and fee instruments, individual exemptions, directions, approvals and permits, definitions-only parts, tables of contents), incorporation decision, language copy, catalogue page, stub (under 150 words without a substantive capture), byte-identical duplicate, or dependent act. Headline counts use principal instruments (principal plus amendment); instrument families group yearly renewals, successive versions and parts of one regime; and two text-based tiers record whether the full text contains autonomy or automation provisions and whether it names artificial intelligence explicitly. The same rules reclassify standards still in development as unpublished, mandatory Chinese national standards as binding, treaty-process working papers as position documents, and proposals whose adopted act is present as superseded.

Doc-level corrections with a named source were taken from the governance expert's audit table: 165 judged rows were converted mechanically into field-level overrides (unit, class, status, year, relevance), each carrying its reason and source, and applied after the rules; a second reading of the expert's fix text against the converted values, prompted by the writers, found eleven inverted values (an in-force directive marked superseded, mandatory standards marked soft) and corrected them, leaving 157 overrides on 91 documents.

Definitions used by every analysis were unified in one loader: a single instrument base (principal instruments), binding counted on that base, a content-capture flag with a minimum of 150 words for any statistic about depth of treatment, and a collection-round flag.

Analyses were extended rather than replaced: the collection round entered the covariate-adjusted models and the trend models; the embodied-versus-general principle contrast was computed within each round and within instruments of each round; regional divergence was computed within research outputs and within principal instruments; odds ratios from sparse cells were flagged as not estimable; the coverage audit was reported under a strict criterion (a principal instrument with a content-analysed capture) and the earlier loose one; and the rule-based unit was compared with the precision audit on the same documents (agreement 0.77, kappa 0.53; the rule keeps 15 documents the auditor rejects and rejects 8 the auditor keeps).

Annotation inputs were repaired by making the cleaner character-aware and re-annotating all 422 Chinese, Japanese and Korean documents under the same policy; 57 changed their embodied-set membership and the number rated "not about AI" fell from 205 to 160.

Figures, dashboard and narratives were then regenerated: computed titles wrapped to the figure, four false titles replaced by computed ones, the regional-similarity figure rebuilt, the dashboard's methods panel and tiles generated from the data with the principal unit, a round-

2 view and robot-type filters, and all six narrative documents rewritten by writer agents from a written brief and checked by a numbers audit.

6. Effect on the findings

On the comparable unit (every text of instrument class) the collection holds 810 embodied-AI instruments and 291 binding ones, against 65 and 5 in version 2; on the version-4 unit it holds 505 principal instruments, 408 of them rated robot-relevant, 195 binding, 168 of those robot-relevant, 140 with autonomy or AI provisions and 15 naming AI explicitly. The precision audit's estimate of about 161 genuine binding instruments in the version-3 set and the rule-based 195 bracket the truth from below and above. The order-of-magnitude rise between versions 2 and 4 is retained as a statement about collection strategy, not about the world. The pooled finding that robot instruments discuss core ethical principles less than general-AI instruments holds within round 2 but not within round 1, where no core principle differs, and is therefore reported as a composition effect of the harvested legal texts; the research-versus-governance gap for companion and care robots and the dominance of drone and vehicle law among binding instruments survive every stratification. Coverage of the reference list is 22 of 39 on the strict criterion and 31 of 39 on the loose one.

7. Limits of the procedure

The reviewers, the collectors and the annotator are all language models; the procedure tests consistency across independent agents and against recomputation, not agreement with human experts, and no human has read or labelled any document. The precision audit judged 100 documents with one reader and no second opinion, so its shares carry the stated intervals and an unmeasured reader effect. Rules written from a risk-stratified audit over-correct some genuine instruments (the rule-versus-audit disagreements are listed) and cannot see whether a text contains an operative robot provision. Consolidated statutes still carry the year the annotation found, and 189 instruments that collectors verified but the execution environment could not read remain outside every count. The cycle is reproducible from the briefs, scripts, override files and manifests in the package, and is designed to be run again on the next version.